LSCF: Long-Term Semantic-Guidance ConvFormer for Referring Remote Sensing Image Segmentation

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***Abstract*— Referring remote sensing image segmentation (RRSIS) task aims to generate segmentation masks for target objects based on language descriptions. It requires precise localization while distinguishing between visually similar yet semantically distinct objects. Fusing vision-language features only during extraction causes information loss and semantic forgetting in the decoder, harming similar target distinction. In addition, high-resolution remote sensing images present challenges, including complex backgrounds, diverse object scales, and intricate boundaries, limiting the effectiveness of previous methods. To address these issues, we propose the long-term semantic-guidance ConvFormer (LSCF) Network. First, we fuse multireceptive-field local features extracted by the multiscale coordconv (MCC) module with language-aware global features from the cross-modal attention (CA) module to obtain multimodal representations. Second, the sampling attention (SA) module enables fine-grained vision context alignment under semantic guidance. Finally, the global language fusion (GLF) module is incorporated in the decoder to maintain long-term vision-language alignment and mitigate semantic degradation. Experimental validation on the RefSegRS, RRSIS-D, and RISBench datasets demonstrates that LSCF achieves overall intersection of union (oIoU) scores of 83.27%, 77.42%, and 74.88%, and mean intersection over union (mIoU) scores of 77.44%, 64.25%, and 68.53%, respectively. On RefSegRS, LSCF surpasses the SOTA method FIANet by 5.53% (oIoU) and 9.58% (mIoU), while delivering competitive performance on RRSIS-D and RISBench. Code and experimental configurations will be released.**

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I. Introduction

With advances in remote sensing technologies, increasing data availability has enabled significant progress in interpretation tasks [1], [2], [3], [4], [5], [6], [7]. Meanwhile, breakthroughs in natural language processing (NLP) have accelerated the development of multimodal remote sensing tasks [2], [8], [9], [10]. Integrating the language information [11], [12], [13] provides additional guidance, expanding applications, and improving performance.

Given a language description such as "vehicle driving on the road" and a corresponding image, the goal of referring remote sensing image segmentation (RRSIS) [9] is to generate a segmentation mask for the "vehicle" object. Unlike conventional vision-only approaches in semantic segmentation [14], [15], [16], [17], object detection [3], [18], [19], [20], [21], [22], and instance segmentation [23], [24], [25], [26], RRSIS breaks the limitation of fixed training categories by incorporating language-guided localization. It enables fine-grained differentiation of multiple targets within the same semantic class. With a knowledgeable RRSIS model, target localization can be achieved through language inputs, offering broad application prospects in environmental monitoring [27], [28] and land use analysis [29].

Referring image segmentation (RIS) task has already seen extensive research in the natural domain, with a primary challenge being the alignment and fusion of vision and language features. Early methods [30], [31], [32] utilize CNN and RNN as feature encoders and simple channel concatenation is applied to obtain the fused features. Subsequent works focus on improving the fusion methods, including hierarchical progressive fusion [33], [34], [35], bidirectional vision-language fusion [36], [37], and graph relation reasoning [38], [39]. As Transformer [11] has shown excellent results in the vision and NLP domains, more recent works [40], [41], [42], [43] have adopted transformers as both vision and language encoders, leveraging cross-attention [40], [43], [44] or bidirectional cross-attention [42], [45], [46], [47] to achieve feature fusion.

In remote sensing, LGCE [9], RMSIN [48], CroBIM [49], and FIANet [50] have introduced a series of improvements

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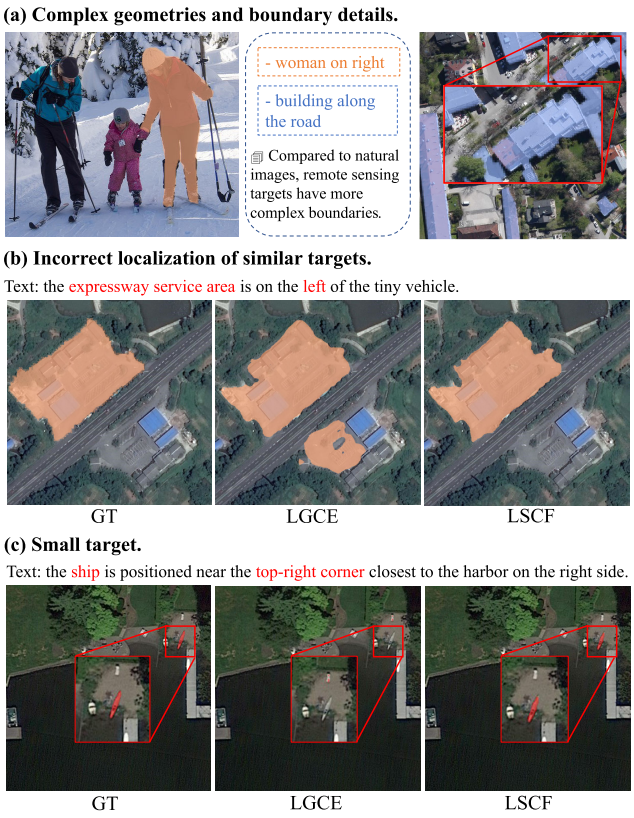


Fig. 1. Three limitations in RRSIS. (a) Remote sensing targets such as "buildings" and "golf courses" often have complex and irregular boundaries. (b) Distinguishing between multiple objects with the same semantic category requires a deeper understanding of contextual linguistic information for accurate target localization. (c) Typically small targets of remote sensing further increase the challenge of precise segmentation.

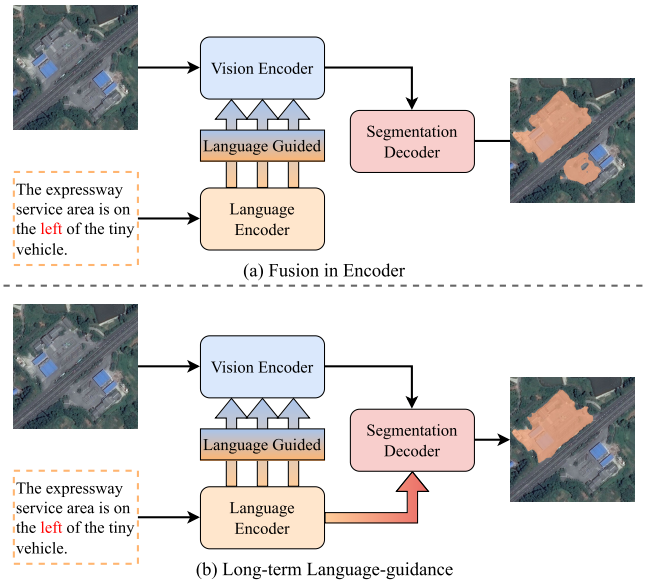


Fig. 2. Conceptual comparison of RRSIS framework. (a) Language is guided only in feature fusion. (b) Long-term language-guidance.

for the RRSIS task. LGCE [9] first applies RIS to remote sensing and establishes the RefSegRS dataset. RMSIN [48] introduces the large-scale, multiresolution RRSIS-D dataset and proposes the cross-scale interaction module (CIM) and adaptive rotated convolution (ARC) to improve multiscale feature fusion and rotated object recognition. CroBIM [49] presents the RISBench dataset and a framework combining the context-aware prompt module (CAPM), language-guided feature aggregation (LGFA), and mutual-interaction decoder (MID) for improved cross-modal segmentation. FIANet [50] improves image–text alignment by employing the fine-grained image–text alignment module (FIAM) to hierarchically fuse global, target related, and positional text.

Despite recent progress, the RRSIS task still faces several key challenges. First, as shown in Fig. 1(a), remote sensing objects often have complex geometries and detailed boundaries [14], [15], [16], making precise edge delineation difficult for conventional methods. Second, as illustrated in Fig. 1(b), language descriptions impose fine-grained identification constraints, requiring an accurate distinction between similar objects within the same category. Although methods like LGCE [9] attempt to improve language alignment, they mainly integrate linguistic features during feature encoding or fusion, while neglecting their role in the decoder stage (see Fig. 2). This limits semantic consistency and segmentation accuracy. Finally, as shown in Fig. 1(c), multiscale variations and small object detection remain unresolved [48], [49], [50]. Due to significant scale diversity in remote sensing imagery, direct multiscale fusion often causes information loss or feature imbalance, impairing small object recognition. Therefore, enhancing semantic guidance throughout the encoder–decoder process and improving the model's ability to handle complex boundaries and small targets remain crucial directions for future RRSIS research.

To address the above challenges, we propose the long-term semantic-guidance ConvFormer (LSCF), as shown in Fig. 3. LSCF comprises two components: the language-aware multiscale ConvFormer (LMCF) block and the cross-scale language-guided alignment (CLA) segmentation head. To alleviate the issues of multiscale variations and small object recognition, the LMCF block adopts a dual-branch structure. The convolutional branch uses the multiscale coordconv (MCC) module to adjust receptive fields via different kernel sizes, improving feature extraction across scales. The transformer branch employs the cross-modal attention (CA) module for vision-language alignment and global context modeling. Both branches incorporate relative positional encoding (CoordConv) to enhance spatial representation. In addition, as the image contains multiple semantically similar objects, the sampling attention (SA) module filters out ambiguous regions, producing fine-grained features with stronger representational capacity. In the CLA head, to mitigate semantic forgetting from stacked convolutions, language features are integrated into the decoder via the global language fusion (GLF) module. However, interpolation-based upsampling often causes intra-class inconsistency and boundary displacement, which weakens the effectiveness of language guidance, so frequency fusion [51] is introduced to better align feature maps and enhance model performance.

In summary, our main contributions are summarized in the following three points.

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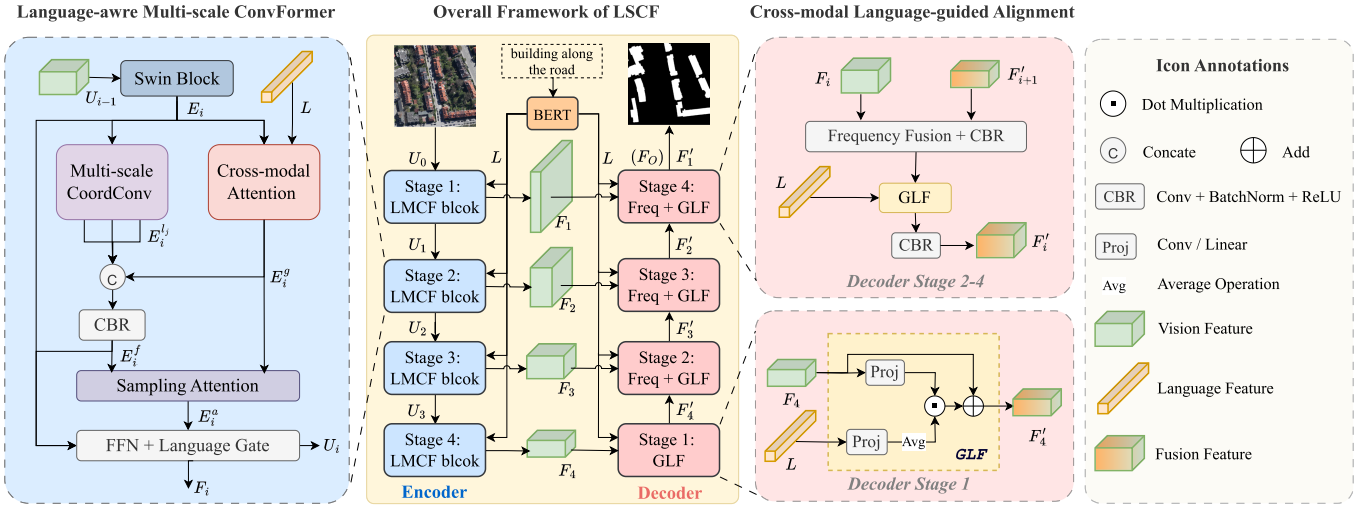


Fig. 3. Overview of LSCF. We utilize Swin Transformer [65] as the vision encoder. At each stage, the vision features E_i, $i \in \{1, 2, 3, 4\}$, is first obtained through the Swin block. E_i is then processed through the MCC to obtain multiscale vision features $E_i^{l_j}$, $j \in \{1, 3, 5\}$, and through the CA to capture language-aware global vision information E_i^g. Next, $E_i^{l_j}$ and E_i^g are input into the SA and FFN to produce F_i, $i \in \{1, 2, 3, 4\}$. The LG mechanism from LAVT [40] is inherited, using U_i, $i \in \{1, 2, 3\}$, as the vision input for the encoder of the next layer. In the decoder stage, F_i is fed layer-by-layer into the CLA Head, which is composed of the GLF and frequency fusion [51], to produce the final prediction F_O.

1) We propose a novel RRSIS method named LSCF, which demonstrates excellent generalization capabilities, achieving SOTA performance on RefSegRS [9] dataset and SOTA-comparable performance on RRSIS-D [48] and RISBench [49] datasets.
2) The LMCF integrates the MCC, CA, and SA modules to enhance spatial awareness, guide global feature extraction, and align vision-language, addressing challenges from multiscale variations and semantically distinct objects.
3) The CLA head introduces the GLF module to prevent semantic forgetting and applies frequency fusion [51] during upsampling to reduce intraclass inconsistency and boundary displacement.

II. Related Works

A. Referring Image Segmentation

RIS was first introduced by Hu et al. [30] in the natural domain, aiming to generate a segmentation mask for a given language description. The standard RIS pipeline involves three main steps.

1) Extracting vision and language features using separate encoders.
2) Aligning and fusing these features via a cross-modal interaction module.
3) Decoding the fused representation to predict the final mask.

Early methods [30], [31], [32], [52], [53], [54], [55] typically used CNNs and RNNs as encoders, combining the features through simple concatenation. Subsequent approaches improved fusion by incorporating hierarchical progressive strategies [33], [34], [35], bidirectional cross-modal interactions [36], [37], and relationship reasoning mechanisms [38], [39]. These enhancements improve contextual understanding between vision and language modalities, thereby boosting segmentation performance.

With the advent of Transformers [11], which have shown remarkable performance in image segmentation [56], [57], object detection [58], [59], and vision-language understanding [60], [61], researchers have shifted toward transformer-based RIS models. VLT [62] and ReSTR [63] were among the first to adopt unified transformer architectures, encoding both images and language descriptions using transformer encoders [13], [64], [65], [66], followed by feature fusion in a transformer decoder. These methods significantly improve global vision-language alignment, outperforming traditional CNN-based approaches. Later works such as LAVT [40] and CEFNet [37] investigated optimal fusion stages and found early stage fusion more effective. LAVT introduces language-aware feature extraction, where the pixelwise attention module (PWAM) and language gate (LG) enhance vision representations at multiple scales using language guidance. Building on this, SLViT [44] improves unidirectional alignment under language supervision for more precise segmentation. Meanwhile, bidirectional strategies such as DMMI [42], CARIS [67], and MagNet [47] enhance early fusion by incorporating hierarchical pixel-language alignment throughout the network.

Beyond architectural improvements, recent efforts have focused on leveraging large-scale pre-trained vision-language models. CRIS [68], GL-CLIP [69], BKINet [70], and Prompt-RIS [71] incorporate CLIP [60] and SAM [72], exploiting pretrained knowledge from large vision-language datasets to improve generalization across diverse domains. In addition, semi-supervised and weakly supervised learning paradigms have gained traction. Methods such as SaG [73], Intra-chunk [74], and ResMatch [75] demonstrate that utilizing unlabeled data effectively enhances segmentation robustness.

However, applying existing RIS methods directly to remote sensing datasets is often ineffective due to inherent challenges such as complex boundaries, multiscale objects, and precise localization. These characteristics demand tailored adaptations

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to achieve accurate segmentation performance in the remote sensing domain.

B. Referring Remote Sensing Image Segmentation

With rapid advances in natural image research, the RIS task has been extended to remote sensing. LGCE [9] first introduced the RRSIS task, along with the RefSegRS dataset. It incorporates a cross-scale feature fusion module to better align high-level vision and language information, addressing challenges posed by small and scattered targets. RMSIN [48] follows with a larger and more complex dataset, RRSIS-D, enriched with diverse semantic information. To handle multiscale details, it introduces the intrascale interaction module (IIM) and CIM and further employs a rotated convolution layer in the decoder to tackle target rotation issues. From the language perspective, FIANet [50] proposes the FIAM, enabling multilevel alignment between vision features and global, target-specific, and positional text. It also integrates language guidance into the cross-scale fusion process. CroBIM [49] presents a cross-modal bidirectional interaction framework, combining CAPM for spatial-aware linguistic enhancement, LGFA for language-guided feature aggregation, and MID for bidirectional alignment, leading to significant improvements in segmentation accuracy.

While LGCE [9], RMSIN [48], FIANet [50], and CroBIM [49] have achieved improved performance on their respective datasets, both methods fail to fully incorporate language features into the decoder head and overlook issues of intraclass inconsistency and boundary displacement in multiscale vision feature fusion.

III. LSCF Method

A. Architecture Overview

The overall architecture of the LSCF is shown in Fig. 3. Swin Transformer [65] and BERT [13] are adopted as the vision and language feature extractors, respectively. Given an image $X \in \mathbb{R}^{H\times W\times 3}$ and a language description $T \in \mathbb{R}^{N}$, where H and W denote the height and width of the image and N denotes the number of words in the text, BERT encodes the text into features $L \in \mathbb{R}^{N\times C_t}$, where C_t is the dimensionality of the language embedding.

The image X and language features L are fed into four LMCF encoder stages to generate multimodal fusion features $F_i \in \mathbb{R}^{H_i\times W_i\times C_i}$, $i \in 1, 2, 3, 4$, where C_i, H_i, and W_i denote the channel, height, and width at each stage, respectively. Each encoder stage first extracts vision features E_i using a Swin Transformer block. These features are then processed by the MCC module to produce local features $E_i^{l_1}$, $E_i^{l_3}$, and $E_i^{l_5}$ at different scales, and by the CA module to obtain language-aware global features E_i^g. The local and global features are concatenated and passed through a convolution-batch norm-ReLU (CBR) block to form fused features E_i^f, which, together with E_i^g, are refined via an SA module and a feed-forward network (FFN) to yield the final fused feature F_i. These multiscale features F_i, along with the language features L, are then input to the CLA decoder. As shown in the pink area of Fig. 3, stage 1 employs a GLF module, while stages 2–4 use frequency fusion [51] for upsampling, followed by semantic integration through GLF. Sections III-B and III-C detail each component of the proposed framework.

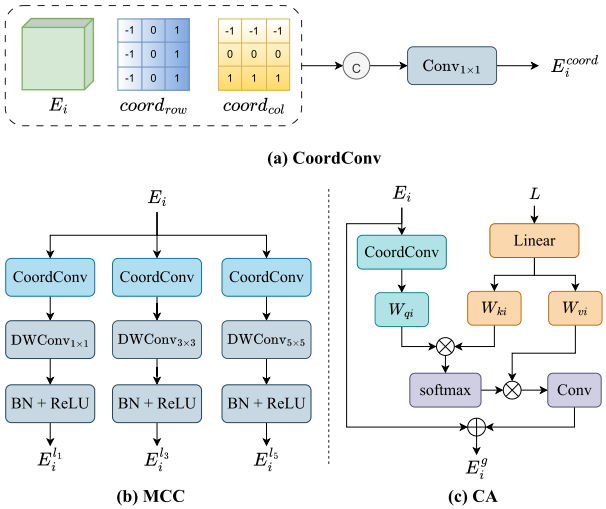


Fig. 4. Pipeline of CoordConv, MCC, and CA. (a) Based on the height and width of the vision features, generate positional coordinate matrices $\text{coord}_{\text{row}}$ and $\text{coord}_{\text{col}}$. Concatenate these matrices and apply convolution to aggregate the information. (b) First, incorporate relative positional information into the vision features E_i by CoordConv, and then obtain multiscale vision features using depthwise separable convolutions with kernel sizes of 1, 3, and 5. (c) Vision features E_i are used as the query, with the language features L serving as the key and value. CoordConv is applied to introduce relative positional information.

B. LMCF Block

Remote sensing targets exhibit multiscale characteristics, and relying solely on transformers may weaken the recognition of small objects. To address this, we introduce a multiscale convolution branch to enhance the receptive fields of local information. Considering the complex geographical relationships inherent in remote sensing images, positional encoding is incorporated into both the convolution and transformer branches, forming the MCC and CA modules, respectively. In addition, an SA module is designed to aggregate local and global features while filtering out ambiguous regions, enabling fine-grained vision context alignment and reducing confusion among visually similar targets.

1) MCC Module: We design MCC to enhance the local spatial information and inductive bias in the vision features, thereby improving the capability to handle small objects in remote sensing. In addition, coordinate positional encoding [CoordConv shown in Fig. 4(a)] is introduced to better understand the spatial relationships between confusing targets. The specific structure of MCC is shown in Fig. 4(b). Based on the vision features E_i, positional coordinates $\text{coord}_{\text{col}}$ and $\text{coord}_{\text{row}}$ are generated as follows:

$$\text{row}_{\text{range}} = \text{linspace}(-1, 1, W_i) \tag{1}$$

$$\text{col}_{\text{range}} = \text{linspace}(-1, 1, H_i) \tag{2}$$

$$\text{coord}_{\text{col}}, \text{coord}_{\text{row}} = \text{meshgrid}(\text{col}_{\text{range}}, \text{row}_{\text{range}}) \tag{3}$$

where H_i and W_i denote the height and width of the input vision feature map, respectively. Then, a 1×1 convolution is applied to integrate the positional information into the

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feature map, followed by depthwise separable convolutions with kernel sizes $j \in \{1, 3, 5\}$ to capture local information

$$E_i^{l_j} = \mathrm{DWConv}_j(\mathrm{Conv}([E_i, \mathrm{coord}_{\mathrm{row}}, \mathrm{coord}_{\mathrm{col}}])). \quad (4)$$

2) CA Module: As shown in Fig. 4(c), we use CA to integrate vision and language features, resulting in language-aware global vision features. Similarly, CoordConv is incorporated into the vision modality to align spatial information between the vision and language features. Based on the vision features E_i and language feature L, first pass L through a feature mapping module for channel alignment, yielding $L' \in \mathbb{R}^{N \times C_i}$

$$L' = \mathrm{GELU}(\mathrm{LN}(L)). \quad (5)$$

The vision features E_i are then treated as the query, with the language feature L' acting as the key and value in the cross-attention calculation, deriving the fused features $E_i^g \in \mathbb{R}^{H_i \times W_i \times C_i}$ according to the following formula:

$$Q_{E_i} = \mathrm{Coord\ Conv}(\mathrm{LN}(E_i))W_{qi} \quad (6)$$

$$K_{L'} = \mathrm{LN}(L')W_{ki} \quad (7)$$

$$V_{L'} = \mathrm{LN}(L')W_{vi} \quad (8)$$

$$E_i^g = E_i + \mathrm{Conv}\left(\mathrm{softmax}\left(\frac{Q_{E_i}^T K_{L'}}{\sqrt{C_i}}\right)V_{L'}^T\right) \quad (9)$$

where LN represents layer normalization and W_{qi}, W_{ki}, and W_{vi} denote the projection functions. Each of the language projection functions W_{ki} and W_{vi} is implemented using a 1×1 convolution. For the vision projection function W_{qi}, a 1×1 convolution is followed by an instance normalization. More delicately, the number of heads is set to 8.

Next, a simple CBR operation is applied to fuse the multiscale local vision features $E_i^{l_j}, j \in \{1, 3, 5\}$ with the language-guided global vision features E_i^g. This produces the fused vision feature $E_i^f \in \mathbb{R}^{H_i \times W_i \times C_i}$, which integrates local representation capability, global representation capability, and semantic representation capability

$$E_i^f = \mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{Conv}\Big(\Big[E_i^{l_j}, E_i^g\Big]\Big)\Big)\Big) \quad (10)$$

where j denotes the kernel size used to extract multiscale local features and conv represents the 1×1 convolutional module with $4 \times C_i$ input channels and C_i output channels.

3) SA Module: Further aggregating the fused features E_i^f is crucial for enhancing fine-grained semantic representation capability. However, directly computing multihead self-attention on E_i^f can blur semantic representations, as complex background information may cause the aggregated features to focus more on unrelated prominent objects. In addition, self-attention within the image itself imposes substantial computational costs.

To address this, we design SA shown in Fig. 5 that samples key regional features from the language-guided global vision feature E_i^g. Specifically, MaxPooling is used to extract regional focal points from E_i^g as the key $K_{E_i^g} \in \mathbb{R}^{H_{pi} \times W_{pi} \times C_i}$, and AvgPooling is used to obtain regional representations from E_i^g as the value $V_{E_i^g} \in \mathbb{R}^{H_{pi} \times W_{pi} \times C_i}$, where H_{p_i} and W_{p_i} represent the height and width of the feature map after pooling operations, respectively. The fused feature E_i^f serves

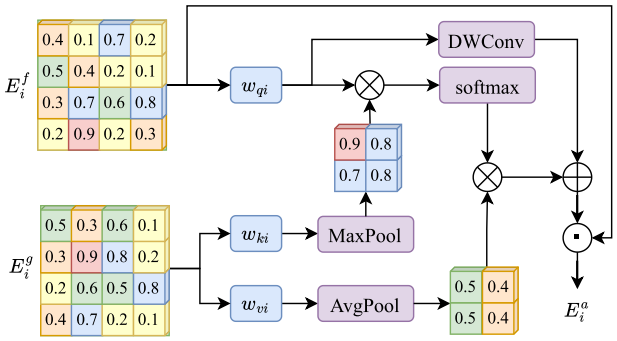


Fig. 5. Diagram of the SA. The global features E_i^g are processed with MaxPool and AvgPool to serve as the key and value, while the fused features E_i^f act as the query. The cross attention is then applied to obtain the aggregated features E_i^a.

as the query to compute similarity with $K_{E_i^g}$. We then assess the attention across different spatial positions to obtain the aggregated feature $E_i^a \in \mathbb{R}^{H_i \times W_i \times C_i}$

$$Q_{E_i^f} = \mathrm{LN}\left(E_i^f\right)w_{qi} \quad (11)$$

$$K_{E_i^g} = \mathrm{MaxPooling}\left(\mathrm{LN}\left(E_i^g\right)w_{ki}\right) \quad (12)$$

$$V_{E_i^g} = \mathrm{AvgPooling}\left(\mathrm{LN}\left(E_i^g\right)w_{vi}\right) \quad (13)$$

$$A_i = \mathrm{softmax}\left(\frac{Q_{E_i^f}^T K_{E_i^g}}{\sqrt{C_i}}\right)V_{E_i^g}^T \quad (14)$$

$$E_i^a = E_i^f \odot w_{oi}\left(\mathrm{DWConv}\left(Q_{E_i^f}\right) + A_i\right) \quad (15)$$

where w_{qi}, w_{ki}, w_{vi}, and w_{oi} represent the projection function implemented by a linear layer. DWConv denotes a depthwise separable convolution with a kernel size of 3×3, while $\odot$ represents the dot product operation. For both MaxPooling and AvgPooling, the window size is set to 16.

To align with [11] and [64], after aggregating the feature E_i^a, the FFN is introduced while preserving the residual connection structure, resulting in the final vision feature F_i, as shown in the following equation:

$$F_i = \left(E_i^a + E_i^f\right) + \mathrm{FFN}\left(\mathrm{LN}\left(E_i^a + E_i^f\right)\right) \quad (16)$$

where the FFN consists of two fully connected layers, with the hidden layer mapped to a dimension of $4 \times C_i$. Similarly, we adopt the language path strategy from LAVT [40], applying the gating mechanism on F_i to produce U_i, which serves as the input for the next Swin-Block

$$U_i = E_i + F_i \odot \mathrm{Linear}(\mathrm{ReLU}(\mathrm{Linear}(\mathrm{Tanh}(F_i)))). \quad (17)$$

Through these steps, we obtain $F_i, i \in \{1, 2, 3, 4\}$ as the vision input for the decoder.

C. Cross-Modal Language-Guided Alignment Head

Through the encoder, language-enhanced multiscale vision features $F_i, i \in 1, 2, 3, 4$ are obtained. Directly feeding F_i into the decoder may cause semantic information loss, particularly for small targets. To mitigate this, the GLF module introduces language cues during decoding. In addition, interpolation-based upsampling often results in intra-class inconsistency and boundary shifts across scales. To address this, frequency fusion [51], followed by concatenation and GLF, is employed in the top–down multiscale decoding process [40].

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1) GLF Module: For the language-enhanced vision feature F_4, only the GLF is applied to further enhance its semantic information, resulting in F_4'. First, both F_4 and L are mapped to the same channel dimension C_d through vision projection and language projection modules. Then, the overall semantic information L_g can be obtained through the average operation. The vision feature Z_4 and language feature L_g are further fused through a simple elementwise multiplication. The specific formulas are as follows:

$$Z_4 = \text{Tanh}(\text{Conv}_{1\times1}(F_4)) \tag{18}$$

$$L_g = \text{Avg}(\text{Tanh}(\text{Linear}(L))) \tag{19}$$

$$F_4' = F_4 + Z_4 \odot L_g. \tag{20}$$

2) Decoder Stages 2–4: For the remaining three stages, F_i and F_{i+1}' are used as vision inputs. First, frequency fusion [51] is applied to perform a $2\times$ upsampling on F_i and F_{i+1}', resulting in F_{i+1}^{up}. Then, F_{i+1}^{up} is concatenated with F_i along the channel dimension and mapped to the decoding embedding dimension C_d, yielding $Z_i, i \in \{3, 2, 1\}$. Next, Z_i and L are input into the GLF followed by a simple CBR module to obtain F_i'. The detailed process is as follows:

$$F_{i+1}^{\text{up}} = \text{Frequency Fusion}\left(F_i, F_{i+1}'\right) \tag{21}$$

$$Z_i = \text{CBR}\left(\left[F_i, F_{i+1}^{\text{up}}\right]\right) \tag{22}$$

$$F_i' = \text{CBR}(\text{GLF}(Z_i, L)). \tag{23}$$

Finally, convolution and upsampling are applied to the output feature F_1' to match the original image size, yielding the final segmentation result F_O.

D. Loss Function

In the RRSIS task, object targets are typically treated as the foreground, while all other pixels are considered as the background, making it a binary classification problem. Due to the category imbalance characterized by small objects, only using the cross-entropy loss may cause the model to overly focus on the background, thereby degrading its ability to detect target regions effectively.

To mitigate this limitation, we integrate Dice loss [76] that directly quantifies mask similarity through IoU-based metrics. Dice loss emphasizes segmentation region integrity, improves boundary precision, and is particularly sensitive to small target areas, ensuring continuous model attention during optimization. Therefore, the combined loss function is adopted as the training objective

$$L = \lambda \cdot L_{\text{CE}}(F_O, Y) + (1-\lambda) \cdot L_{\text{DICE}}(F_O, Y) \tag{24}$$

where λ is a hyperparameter balancing the two loss functions, set to 0.9 in this study. $F_O, Y \in \mathbb{R}^{H\times W}$ represents the upsampled prediction and the ground-truth mask, respectively. In addition, we assign different weights to the foreground and background in the cross-entropy loss, setting a ratio of 1.1:0.9 to further mitigate class imbalance.

IV. Experiments

A. Datasets and Metrics

We comprehensively assess LSCF on three benchmark datasets: RefSegRS [9], RRSIS-D [48], and RISBench [49], demonstrating its effectiveness and robustness in remote sensing image segmentation.

RefSegRS [9] is the first dataset for RRSIS, containing 2172 training, 431 validation, and 1817 testing samples. All images are standardized to 512×512 pixels with a spatial resolution of 0.13 m. Derived from SkyScapes [77], it uses three templates to generate referring expressions covering 14 semantic categories, including road, vehicle, building, and tree. *RRSIS-D* [9] provides 17 402 image–text pairs at 800×800 resolution, with 20 semantic categories and seven attribute types. The data are split into 12 181 training, 1740 validation, and 3481 testing samples. It is built upon RSVGD [10], with object masks generated via SAM [72] from bounding box annotations and manually refined. *RISBench* [49], based on VRSBench [78], includes 52 472 image-language-mask triplets at 512×512 pixels, spanning spatial resolutions from 0.1 to 30 m. It covers 26 semantic categories, each with eight attributes, and is divided into 26 300 training, 10 013 validation, and 16 149 testing samples.

In line with prior work [9], [40], [48], [49], [50], we use oIoU as the primary evaluation metric, with mIoU and Precision@X (Pr@X, $X = 0.5, 0.6, 0.7, 0.8$, and 0.9) as secondary metrics to comprehensively assess model performance. The oIoU measures the ratio of the total intersection area to the total union area across all testing samples, favoring large objects

$$\text{oIoU} = \frac{\sum_{i=1}^{M} \left|Y^i \cap F_O^i\right|}{\sum_{i=1}^{M} \left|Y^i \cup F_O^i\right|} \tag{25}$$

where Y^i denotes the ground-truth mask of the i sample, F_O^i represents the predicted mask, $|\cdot|$ indicates the pixelwise area, and M is the total number of testing samples. Meanwhile, the mIoU is the average IoU computed for each sample individually, which provides a holistic performance measure and treats objects of all sizes equally

$$\text{mIoU} = \frac{1}{M} \sum_{i=1}^{M} \frac{\left|Y^i \cap F_O^i\right|}{\left|Y^i \cup F_O^i\right|}. \tag{26}$$

Pr@X indicates the proportion of samples where IoU exceeds X, gradually reflecting the closeness between the mask and ground truth

$$\text{Pr@X} = \frac{1}{M} \sum_{i=1}^{M} \mathbb{I}\left(\frac{\left|Y^i \cap F_O^i\right|}{\left|Y^i \cup F_O^i\right|} > X\right) \tag{27}$$

where $\mathbb{I}(\cdot)$ is an indicator function that returns 1 if the condition is satisfied and 0 otherwise. For all these metrics, higher values indicate better performance.

B. Implementation Details

In experiments, the Swin-Transformer-Base [65] pre-trained on ImageNet-22k [79] is used as the vision backbone, while the BERT-Base [13] from HuggingFace [80] is employed as the language backbone. For all three datasets, training is conducted on the training set, and scores are reported on the validation and testing sets. Images are resized to 512×512 and trained with a 3e-5 learning rate, AdamW

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TABLE I

COMPARATIVE EXPERIMENTAL RESULTS ON THE REFSEGRS VALIDATION AND TESTING DATASET [9] WITH SOTA METHODS. THE BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN BOLD. THE EXPERIMENTAL RESULTS ARE QUOTED FROM [49]. † DENOTES THE RESULT OF OUR REPLICATION

Methods	Vision	Language	Pr@5		Pr@6		Pr@7		Pr@8		Pr@9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
BRINet [36]	R-101	LSTM	36.86	20.72	35.53	14.26	19.93	9.87	10.66	2.98	2.84	1.14	61.59	58.22	38.73	31.51
LSCM [38]	R-101	LSTM	56.82	31.54	41.24	20.41	21.85	9.51	12.11	5.29	2.51	0.84	62.82	61.27	40.59	35.54
CMPC [33]	R-101	LSTM	46.09	32.36	26.45	14.14	12.76	6.55	7.42	1.76	1.39	0.22	63.55	55.39	42.08	40.63
CMSA [52]	R-101	None	39.24	28.07	38.44	20.25	20.39	12.71	11.79	5.61	1.52	0.83	65.84	64.53	43.62	41.47
RRN [53]	R-101	LSTM	55.43	30.26	42.98	23.01	23.11	14.87	13.72	7.17	2.64	0.98	69.24	65.06	50.81	41.88
CMPC+ [54]	R-101	LSTM	56.84	49.19	37.59	28.31	20.42	15.31	10.67	8.12	2.78	2.55	70.62	66.53	47.13	43.65
MAttNet [55]	R-101	LSTM	48.56	28.79	40.26	22.51	20.59	11.32	12.98	3.62	2.02	0.79	66.84	64.28	41.73	33.42
CRIS [68]	R-101	CLIP	53.13	35.77	36.19	24.11	24.36	14.36	11.83	6.38	2.55	1.21	72.14	65.87	53.74	43.26
BKINet [70]	R-101	CLIP	52.04	36.12	35.31	20.62	18.35	15.22	12.78	6.26	1.23	1.33	75.37	63.37	56.12	40.41
ETRIS [45]	R-101	CLIP	54.99	35.77	35.03	23.00	25.06	13.98	12.53	6.44	1.62	1.10	72.89	65.96	54.03	43.11
CARIS [67]	Swin-B	BERT	68.45	45.40	47.10	27.19	25.52	15.08	14.62	8.87	3.71	1.98	75.79	69.74	54.30	42.66
RefSegformer [43]	Swin-B	BERT	81.67	50.25	52.44	28.01	30.86	17.83	17.17	9.19	5.80	2.48	77.74	71.13	60.44	47.12
CrossVLT [46]	Swin-B	BERT	67.52	41.94	43.85	25.43	25.99	15.19	14.62	3.71	1.87	1.76	76.12	69.73	55.27	42.81
DMMI [42]	Swin-B	BERT	86.17	63.89	74.71	44.30	38.05	19.81	18.10	6.49	3.25	1.00	74.02	68.58	65.72	52.15
LAVT [40] †	Swin-B	BERT	82.27	63.37	79.40	48.73	59.26	27.17	26.85	12.71	9.03	3.69	82.76	75.04	69.06	52.89
LGCE [9] †	Swin-B	BERT	92.13	74.04	88.66	61.39	77.78	39.71	37.27	17.71	13.19	5.17	85.52	77.71	74.62	59.59
RMSIN [48] †	Swin-B	BERT	90.28	71.07	84.95	57.26	73.15	34.87	36.34	13.15	8.10	2.75	79.65	72.68	73.04	57.91
FIANet [50] †	Swin-B	BERT	95.36	83.60	93.97	75.18	88.63	59.55	69.14	30.65	12.99	6.16	85.17	77.74	80.19	67.86
CroBIM [49]	Swin-B	BERT	87.24	64.83	75.17	44.41	44.78	17.28	19.03	9.69	6.26	2.20	78.85	72.30	65.79	52.69
LSCF	Swin-B	BERT	**97.22**	**87.51**	**96.30**	**82.89**	**93.75**	**75.85**	**89.58**	**62.65**	**72.92**	**33.77**	**90.80**	**83.27**	**89.45**	**77.44**

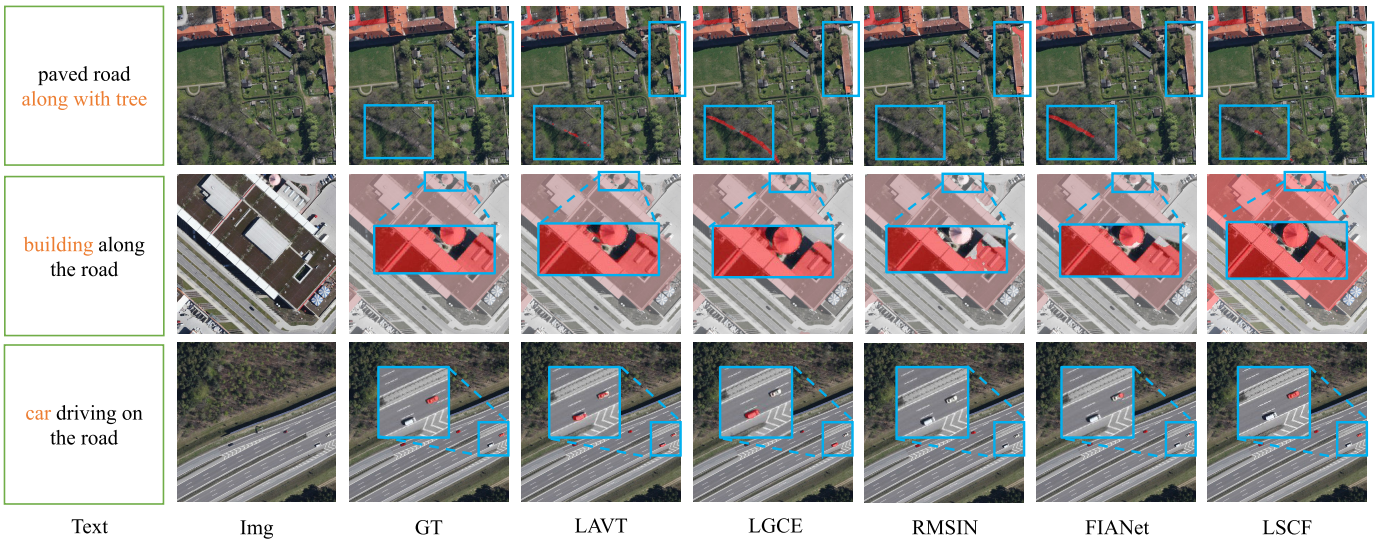


Fig. 6. Comparison of visualization results for RefSegRS dataset. Highlights are marked by blue boxes.

optimizer [81] with 0.01 weight decay, batch size of 8. For the RefSegRS [9] and RISBench [49] datasets, training is conducted for 50 epochs, while the RRSIS-D dataset is trained for 40 epochs. All experiments run on 2 RTX 4090 GPUs in PyTorch [82], ensuring consistent configurations for a fair comparison.

C. Comparison With Other Methods

We compare LSCF with classical and leading methods in RIS and RRSIS, including RRN [53], CMSA [52], LSCM [38], CMPC [33], BRINet [36], CMPC+ [54], MAttNet [55], BKINet [70], ETRIS [45], CRIS [68], LAVT [40], DMMI [42], CrossVLT [46], RefSegformer [43], CARIS [67], LGCE [9], RMSIN [48], FIANet [50], and CroBIM [49]. These approaches encompass early stage methods in natural image processing utilizing ResNet and LSTM architectures, CLIP pretraining-based transfer learning techniques, and advanced methods leveraging Swin Transformer and BERT. In addition, missing experiments for open-source RRSIS methods are supplemented on certain datasets. The experimental results on the three datasets are presented separately below.

1) RefSegRS Dataset: The comparative results of each evaluation metric are presented in Table I. LSCF achieves significant improvements over existing methods and sets a new SOTA. Compared with the natural image baseline LAVT [40], LSCF improves oIoU by 8.23% and mIoU by 24.55% on the testing set. Compared with the remote sensing SOTA method FIANet [50], it improves oIoU by 5.53% and mIoU by 9.58%. These results confirm that LSCF consistently enhances segmentation accuracy across all targets. In particular, the Pr9 metric shows a large gain. On the validation set, LSCF improves by 59.73% over FIANet [50]. On the testing set, it improves by 27.61%. This demonstrates strong performance in handling small object boundaries and verifies the effectiveness of MCC and GLF. Qualitative comparisons are shown

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TABLE II

COMPARISON WITH SOTA METHODS ON RRSIS-D VALIDATION AND TESTING DATASET [9]. THE BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN BOLD. THE EXPERIMENTAL RESULTS ARE QUOTED FROM [49]. † DENOTES THE RESULT OF OUR REPLICATION

Method	Vision	Language	Pr@5		Pr@6		Pr@7		Pr@8		Pr@9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
BRINet [36]	R-101	LSTM	58.79	56.90	49.54	48.77	39.65	39.12	28.21	27.03	9.19	8.73	70.73	69.88	51.14	49.65
LSCM [38]	R-101	LSTM	57.12	56.02	48.04	46.25	37.87	37.70	26.37	25.28	7.93	8.27	69.28	69.05	50.36	49.92
CMPC [33]	R-101	LSTM	57.93	55.83	48.85	47.40	38.50	36.94	25.28	25.45	9.31	9.19	70.15	69.22	50.41	49.24
CMSA [52]	R-101	None	55.68	55.32	48.04	46.45	38.27	37.43	26.55	25.39	9.02	8.15	69.68	69.39	48.85	48.54
RRN [53]	R-101	LSTM	51.09	51.07	42.47	42.11	33.04	32.74	20.80	21.57	6.14	6.37	66.53	66.43	46.06	45.64
CMPC+ [54]	R-101	LSTM	59.19	57.65	49.36	47.51	38.67	36.97	25.91	24.33	8.16	7.78	70.14	68.64	51.41	50.24
CRIS [68]	R-101	CLIP	56.44	54.84	47.87	46.77	39.77	38.06	29.31	28.15	11.84	11.52	70.08	70.46	50.75	49.69
BKINet [70]	R-101	CLIP	58.79	56.90	49.54	48.77	39.65	39.12	28.21	27.03	9.19	9.16	70.78	69.89	51.14	49.65
ETRIS [45]	R-101	CLIP	62.10	61.07	53.73	50.99	43.12	40.94	30.79	29.30	12.90	11.43	72.75	71.06	55.21	54.21
CARIS [67]	Swin-B	BERT	71.61	71.50	64.66	63.52	54.14	52.92	42.76	40.94	23.79	23.90	77.48	77.17	62.88	62.17
RefSegformer [43]	Swin-B	BERT	64.22	66.59	58.72	59.58	50.00	49.93	35.78	33.78	24.31	23.30	76.39	77.40	58.92	58.99
CrossVLT [46]	Swin-B	BERT	67.07	66.42	59.54	59.41	50.80	49.76	40.57	38.67	23.51	23.30	76.25	75.48	59.78	58.48
DMMI [42]	Swin-B	BERT	70.40	68.74	63.05	60.96	54.14	50.33	41.95	38.38	23.85	21.63	77.01	76.20	60.72	60.12
LAVT [40] †	Swin-B	BERT	70.23	69.67	62.41	63.87	54.02	54.14	42.87	42.02	24.69	24.96	77.74	**77.66**	61.73	61.25
LGCE [9] †	Swin-B	BERT	70.80	69.90	63.85	63.96	54.37	53.56	42.76	40.81	24.48	24.21	77.85	77.23	61.94	60.91
RMSIN [48] †	Swin-B	BERT	74.25	74.15	66.78	67.08	55.98	55.64	43.39	41.65	23.22	23.07	76.78	77.49	64.33	63.95
FIANet [50] †	Swin-B	BERT	75.13	74.46	67.45	66.96	56.95	56.31	44.37	42.83	24.48	24.13	77.53	76.91	64.97	64.01
CroBIM [49]	Swin-B	BERT	74.20	**75.00**	66.15	66.32	54.08	54.31	41.38	41.09	22.30	21.78	76.24	76.37	63.99	64.24
LSCF	Swin-B	BERT	**75.17**	74.30	**67.93**	**67.69**	**57.99**	**56.32**	**44.94**	**43.08**	**25.98**	**25.67**	**78.14**	77.42	**65.15**	**64.25**

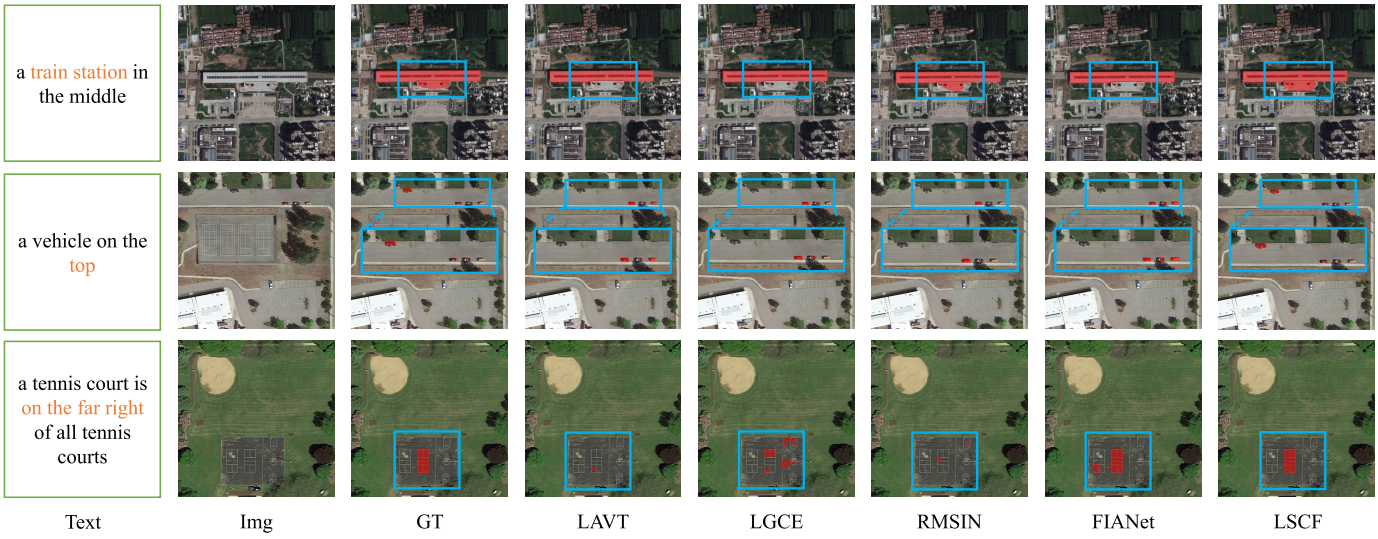


Fig. 7. Comparison of visualization results for RRSIS-D dataset. Highlights are marked by blue boxes.

in Fig. 6. In the second row, LSCF provides more accurate boundary segmentation for the "building." In the third row, it shows a better understanding of the "car" category and distinguishes it more clearly from similar objects. This leads to more precise localization and fewer false positives.

2) RRSIS-D Dataset: Table II summarizes the comparative results on the validation and testing sets. The proposed LSCF framework achieves SOTA performance across multiple metrics. On the validation set, it improves oIoU by 0.29% and mIoU by 0.18% over the previous SOTA. More notable gains appear in Precision@X metrics. Specifically, LSCF surpasses prior methods by 1.04%, 0.55%, and 1.29% at Pr@0.7, Pr@0.8, and Pr@0.9 on the validation set, respectively. On the testing set, it shows improvements of 0.61%, 0.25%, and 0.71% at Pr@0.6, Pr@0.8, and Pr@0.9. In aggregated metrics, LSCF consistently outperforms all individual methods. Qualitative results in Fig. 7 further demonstrate its advantages. In Row 1, LSCF achieves more complete segmentation of the "train station" with clearer boundaries, benefiting from vision-language coherence. In Row 3, it accurately distinguishes ambiguous targets, such as the "far-right tennis court," showing its effectiveness in handling fine-grained semantic distinctions. The findings provide additional evidence for the effectiveness of the SA module.

3) RISBench Dataset: The comparative results are presented in Table III, further confirming the generalization capability and robustness of LSCF. Although LSCF does not achieve the best performance in oIoU, it shows significant improvements in Pr@X and mIoU metrics. Compared with CroBIM [49], LSCF achieves gains of 0.26% at Pr@0.7, **1.74%** at Pr@0.8, 7.85% at Pr@0.9, and 0.36% in mIoU on the validation set. On the testing set, the improvements are **1.84%** at Pr@0.7, **4.01%** at Pr@0.8, **8.28%** at Pr@0.9, and **1.21%** in mIoU. These results demonstrate the model's enhanced ability to localize small objects and refine segmentation boundaries. Qualitative comparisons in Fig. 8 offer further

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TABLE III

COMPARATIVE EXPERIMENTAL RESULTS ON THE RISBENCH VALIDATION AND TESTING DATASET [9] WITH SOTA METHODS. THE BEST RESULTS FOR EACH METRIC ARE HIGHLIGHTED IN BOLD. THE EXPERIMENTAL RESULTS ARE QUOTED FROM [49]. † DENOTES THE RESULT OF OUR REPLICATION

Methods	Vision	Language	Pr@5		Pr@6		Pr@7		Pr@8		Pr@9		oIoU		mIoU	
			Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test	Val	Test
RRN [53]	R-101	LSTM	54.62	55.04	46.88	47.31	39.57	39.86	32.64	32.58	11.57	13.24	47.28	49.67	42.65	43.18
LSCM [38]	R-101	LSTM	55.87	55.26	47.24	47.14	40.22	40.10	33.55	33.29	12.78	13.91	47.99	50.08	43.21	43.69
BRINet [36]	R-101	LSTM	52.11	52.87	45.17	45.39	37.98	38.64	30.88	30.79	10.28	11.86	46.27	48.73	41.54	42.91
MAttNet [55]	R-101	LSTM	56.77	56.83	48.51	48.02	41.53	41.75	34.33	34.18	13.84	15.26	48.66	51.24	44.28	45.71
CMPC [33]	R-101	LSTM	54.89	55.17	47.77	47.84	40.38	40.28	32.89	32.87	12.63	14.55	47.59	50.24	42.83	43.82
CMPC+ [54]	R-101	LSTM	57.84	58.02	49.24	49.00	42.34	42.53	35.77	35.26	14.55	17.88	50.29	53.98	45.81	46.73
ETRIS [45]	R-101	CLIP	59.87	60.98	49.91	51.88	35.88	39.87	20.10	24.49	8.54	11.18	64.09	67.61	51.13	53.06
CRIS [68]	R-101	CLIP	63.42	63.67	54.32	55.73	41.15	44.42	24.66	28.80	10.27	13.27	66.26	69.11	53.64	55.18
CrossVLT [46]	Swin-B	BERT	70.05	70.62	64.29	65.05	56.97	57.40	44.49	45.80	21.47	26.10	69.77	74.33	61.54	62.84
CARIS [67]	Swin-B	BERT	73.46	73.94	68.51	68.93	60.92	62.08	48.47	50.31	24.98	29.08	70.55	**75.10**	64.40	65.79
DMMI [42]	Swin-B	BERT	71.27	72.05	66.02	66.48	58.22	59.07	45.57	47.16	22.43	26.57	**70.58**	74.82	62.62	63.93
RefSegformer [43]	Swin-B	BERT	67.42	69.15	61.72	63.24	53.64	55.33	40.71	43.27	19.43	24.20	69.50	74.23	59.37	61.25
LAVT [40] †	Swin-B	BERT	70.43	71.33	65.73	66.32	58.56	59.37	47.78	48.64	25.94	28.84	69.85	74.71	62.62	63.78
RMSIN [48] †	Swin-B	BERT	74.82	74.54	69.82	69.51	62.98	62.32	51.14	51.53	28.25	30.58	69.49	73.35	66.42	66.86
LGCE [9] †	Swin-B	BERT	70.58	71.19	65.72	65.92	58.60	58.94	46.87	47.87	24.68	27.86	70.55	74.34	62.39	63.39
FIANet [50] †	Swin-B	BERT	74.66	74.80	70.21	69.86	63.41	63.16	52.49	52.70	29.51	31.87	70.17	74.23	66.16	66.83
CroBIM [49]	Swin-B	BERT	**76.59**	75.75	**71.73**	70.34	64.32	63.12	53.18	51.12	28.53	28.45	69.08	73.61	67.52	67.32
LSCF	Swin-B	BERT	75.67	**76.08**	70.99	**71.29**	**64.58**	**64.96**	**54.92**	**55.13**	**36.38**	**36.73**	69.93	74.88	**67.88**	**68.53**

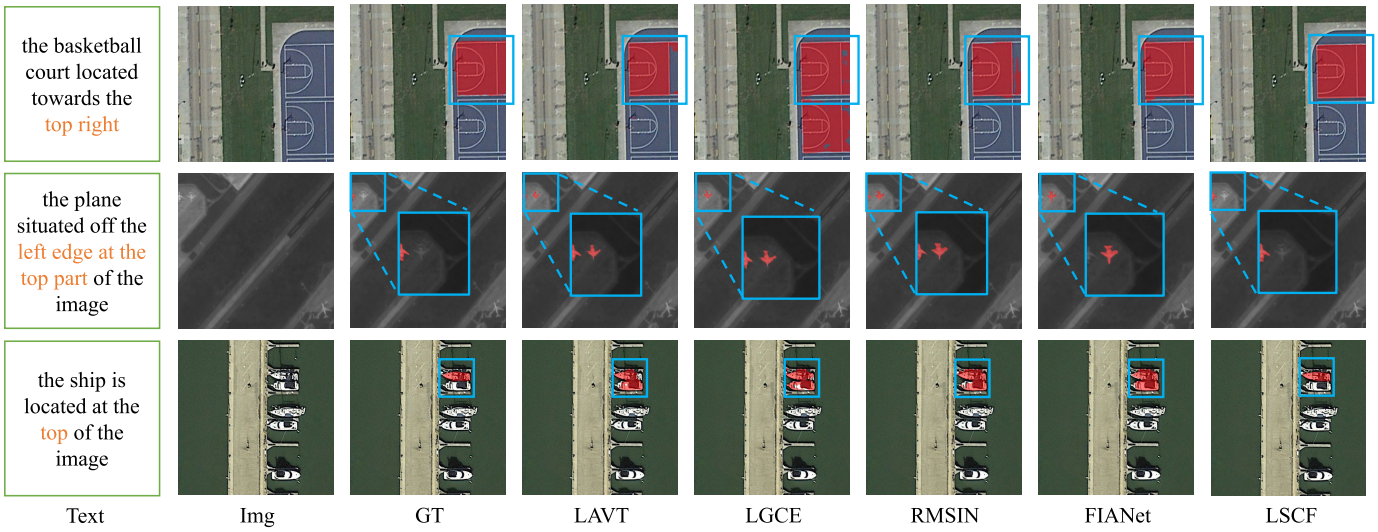


Fig. 8. Comparison of visualization results for RISBench dataset. Highlights are marked by blue boxes.

evidence. In Row 1, LSCF makes better use of vision-language context and generates segmentation masks with sharper edges and fewer artifacts. Rows 2 and 3 show that LSCF performs better in identifying small targets and suppressing interference from semantically similar objects.

D. Ablation Study

We conduct systematic ablation studies on the RefSegRS dataset [9] to validate the efficacy of each proposed module. To improve efficiency, Swin Transformer Tiny [65] is adopted as the vision backbone. The training and validation sets are merged for training, and evaluation is conducted solely on the original testing set, while all other configurations remain identical to the baseline setup on RefSegRS.

1) LMCF Block and CLA Head: To validate the effectiveness of the LMCF block and CLA head, ablation studies are performed on all possible combinations of these two components, as shown in Table IV. Compared with the baseline model LAVT [40], the LMCF block demonstrates improved performance over the PWAM approach, with notable gains in metrics including Pr@5, Pr@7, Pr@9, oIoU, and mIoU. The addition of multiscale convolution strengthens the model's

TABLE IV

ABLATION OF LMCF BLOCK AND CLA HEAD. DEFAULT DENOTES LAVT [40]. ✓INDICATES CORRESPONDING MODULE HAS BEEN REPLACED

LMCF block	CLA head	Pr@5	Pr@7	Pr@9	oIoU	mIoU
Default		78.96	51.15	7.25	79.98	63.92
✓		79.73	56.92	9.01	80.87	65.67
	✓	82.53	60.00	10.66	81.95	67.54
✓	✓	**86.21**	**72.53**	**22.64**	**84.39**	**73.93**

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TABLE V

ABLATION OF MCC, SA, AND COORDCONV MODULES. PWAM DENOTES FUSION MODULE OF LAVT [40] WITH CLA HEAD. † DENOTES CA FUSION WITH CLA HEAD. * INDICATES THE REMOVAL OF COORDCONV

Module	Pr@5	Pr@7	Pr@9	oIoU	mIoU
PWAM	82.53	60.00	10.66	81.95	67.54
+MCC †	85.49	69.34	18.63	83.32	72.41
+SA †	85.05	69.23	16.65	83.53	71.98
LSCF	**86.21**	**72.53**	**22.64**	**84.39**	**73.93**
LSCF*	86.15	71.87	20.16	**84.39**	73.49

TABLE VI

ABLATION OF GLF AND FREQUENCY FUSION [51]. DEFAULT USES LMCF BLOCK AND SEGMENTATION HEAD FROM LAVT [40]

GLF	Freq	Pr@5	Pr@7	Pr@9	oIoU	mIoU
Default		79.73	56.92	9.01	80.87	65.67
✓		84.01	64.78	10.99	82.26	69.98
	✓	81.92	62.80	10.71	81.88	68.01
✓	✓	**86.21**	**72.53**	**22.64**	**84.39**	**73.93**

ability to capture local details, improving small object recognition accuracy. The proposed CLA head introduces language guidance during decoding, enhancing alignment across different feature scales and achieving an **8.85%** improvement in Pr@7. Using both modules together yields significant performance enhancements, particularly in Pr@7, Pr@9, and mIoU scores. These findings affirm the overall design effectiveness of LSCF, especially in the accuracy of segmentation boundaries.

2) LMCF Block: We design three main fusion modules: MCC, SA, and CoordConv. Table V shows their effectiveness. PWAM fusion of LAVT [40] with CLA head is used as the baseline, then each component is added to evaluate its impact. Adding MCC alone boosts Pr@5, Pr@7, Pr@9, oIoU, and mIoU by 2.96%, 9.34%, 7.97%, 1.37%, and 4.87%, respectively, demonstrating the value of multiscale local analysis for detecting small objects. Adding only SA improves Pr@5, Pr@7, Pr@9, oIoU, and mIoU by 2.52%, 9.23%, 5.99%, 1.58%, and 4.44%, showing that local feature sample helps reduce target confusion. When MCC and SA are combined to form the proposed LSCF, multiscale vision features are enhanced with semantic information, achieving the best metrics. Removing CoordConv slightly decreases Pr@5, Pr@7, Pr@9, and mIoU, confirming CoordConv's effectiveness in understanding complex spatial relationships.

3) CLA Head: As shown in Table VI, we explore the structural design of the decoder. Compared to the LAVT [40] segmentation head, adding the GLF alone improves oIoU and mIoU by **1.39%** and **4.31%**, respectively, while introducing frequency fusion [51] alone increases oIoU and mIoU by 1.01% and 2.44%, respectively. The effect of GLF is particularly notable, indicating that introducing language in the decoder head effectively mitigates semantic loss. Building on this, adding frequency fusion for alignment across different scale features further enhances the impact of semantic guidance, resulting in respective improvements of **11.63%** and **11.93%** in the Pr@9 metric.

TABLE VII

ABLATION ON THE NUMBER OF ATTENTION HEADS ACROSS THE FOUR ENCODER STAGES IN LMCF BLOCK

HeadsNum	Pr@5	Pr@7	Pr@9	oIoU	mIoU
1, 1, 1, 1	85.99	71.21	18.24	83.76	72.88
2, 2, 2, 2	85.99	70.93	19.51	83.72	72.86
4, 4, 4, 4	85.49	70.38	21.98	84.15	73.31
8, 8, 8, 8	**86.21**	**72.53**	**22.64**	**84.39**	**73.93**
8, 4, 2, 1	83.52	65.99	12.36	82.40	69.96

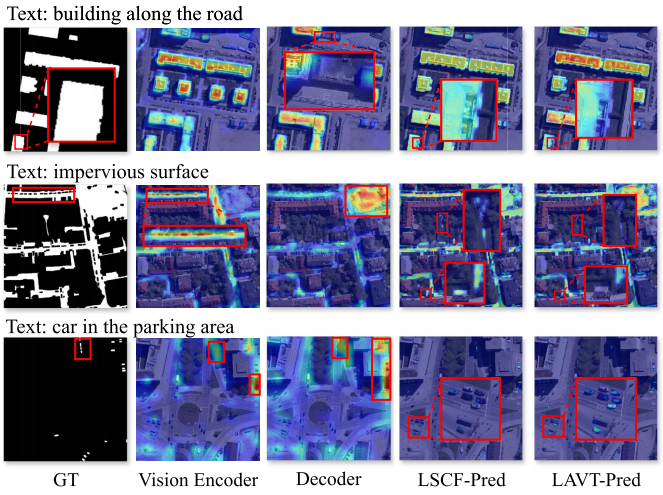


Fig. 9. Attention maps for LSCF and LAVT [40]. LSCF effectively alleviates issues of target confusion, small object recognition, and complex boundary displacement.

4) Heads Number of LMCF Block: We assess the effect of varying attention head counts in the LMCF block. Table VII shows that the {8, 8, 8, 8} configuration performs best. Consistent head numbers generally produce stable results, while the {8, 4, 2, 1} setup results in lower metrics, likely due to the complexity of remote sensing imagery.

5) Qualitative: In Fig. 9, we use GradCAM [83] to visualize the attention distribution across the LSCF encoder–decoder, as well as the prediction attention results of both LSCF and LAVT. For the sample in the first row, the LMCF block accurately localizes all buildings with finer edge details. Compared with LAVT's predictions, LSCF exhibits superior segmentation performance on complex boundaries, verifying the effectiveness of the introduced Frequency Fusion. In the second-row sample, LSCF captures details between impervious surfaces and vehicles during feature extraction, while the GLF in the CLA head incorporates semantics to address partially missed targets. The prediction results further indicate that MCC enhances the sensitivity to small targets. In the third-row sample, LSCF effectively resolves semantic confusion. During feature extraction, LSCF precisely focuses on the parking area, efficiently filtering irrelevant regions and maintaining semantic focus at each stage. This approach minimizes confusion with visually similar objects, such as moving cars on the road, highlighting the importance of the

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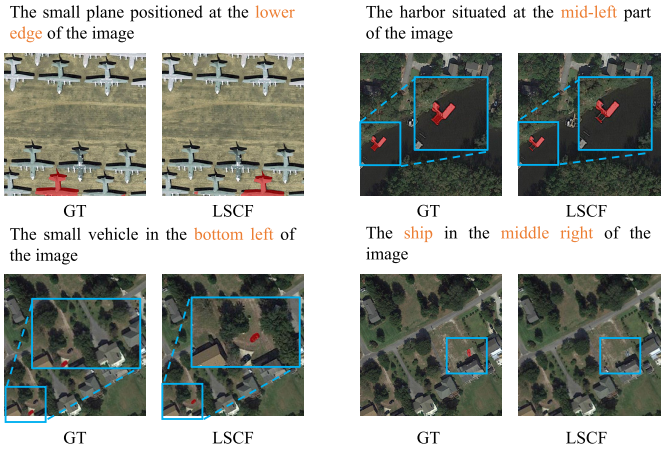


Fig. 10. Failure case results of LSCF.

SA module and the semantic guidance provided by GLF in the decoder head.

V. Discussion

Although LSCF achieves SOTA results on all 3 datasets, some failure cases remain. We select four examples in Fig. 10 and briefly analyze the causes and potential improvements.

In the top-left case, LSCF fails to locate the "plane" due to vague text. The phrase "lower edge" refers to multiple targets, making it hard to identify the correct one. This highlights a common challenge in remote sensing, where many similar targets appear, and simple spatial terms are often ambiguous. The bottom-left case shows a similar issue. While humans can easily distinguish the correct object, the model struggles with understanding fine-grained spatial relationships. In the top-right case, the model partially segments the "harbor" due to unclear semantics. It is uncertain whether docked ships should be included, pointing to the need for better handling of vague terms. The bottom-right case shows a missed "ship" on land. The model likely overlooks it due to context bias, having mostly seen ships in water scenes. Adapting how context is used may help reduce such errors.

VI. Conclusion

In this article, we propose LSCF, a new method for RRSIS that achieves strong performance on RefSegRS and competitive results on RRSIS-D and RISBench. For encoding, MCC captures local features, while CA introduces language semantics and global context. SA fuses multiscale features effectively. In decoding, the GLF module enhances semantics, and frequency fusion improves upsampling. Together, these components enable LSCF to achieve state-of-the-art results.

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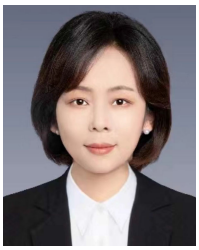
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